

RSA Cryptanalysis: An Elitist Genetic Algorithm for Integer Factorization

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Abstract—The problem of Integer Factorization deals with the disintegration of an integer into prime factors. This has direct applications in cryptanalysis, particularly in breaking the RSA. Lately, scientists have investigated the scope of employing metaheuristic approaches to perform integer factorization in bounded time. This paper improves the state-of-the-art by incorporating the concept of elitism, removal of redundancy, as well as diminishing the effect of exploitation in a genetic algorithm (GA) model. This enabled us to factor integers up to 22 decimal digits (≈ 70 bits) with a higher rate of success and quicker convergence on a standard desktop, thereby breaking the bottleneck of existing metaheuristic approaches. By design, the proposed metaheuristic approach is population based and can process the intermediate steps by means of independent calculations. Hence, it can be easily parallelized and scaled up for factorizing larger integers.

Index Terms—Genetic Algorithm, Integer Factorization, Cryptanalysis, RSA.

I. INTRODUCTION

Cryptanalysis is the study of analyzing and breaking cryptographic systems, aiming to uncover hidden information without access to the secret key [1], [2]. This field combines mathematics, computer science, and engineering to decipher encrypted data, exposing vulnerabilities in cryptographic algorithms. Methods include brute force attacks, where every possible key is tried, and more sophisticated techniques like frequency analysis and differential cryptanalysis, which exploit specific structural weaknesses [3], [4].

Cryptanalysis of RSA involves the study and methods to break the RSA cipher, which is extensively used for secure data transmission [5]. RSA (named after Rivest, Shamir and Adleman) is a foundational asymmetric cryptographic algorithm introduced in 1978 [6]. This algorithm revolutionized secure communication by enabling public-key encryption and digital signatures, ensuring both confidentiality and authenticity. RSA encryption relies on the mathematical difficulty of

factoring large composite numbers, specifically the product of two large prime numbers. The security of RSA is based on the assumption that while it is computationally easy to multiply a pair of primes, it is exceedingly difficult to factor their product back into the original primes, making it a cornerstone of modern secure communication protocols [6]. Cryptanalysts endeavor to identify vulnerabilities or devise algorithms capable of efficiently solving this factoring problem, thereby compromising the security of RSA.

Metaheuristics are high-level problem-independent algorithmic frameworks that provide a set of guidelines or strategies to develop heuristic optimization algorithms [7]. These approaches, which include genetic algorithms (GAs) [8], simulated annealing [9], particle swarm optimization [10], and ant colony optimization [11], etc. offer significant potential in cryptanalysis due to their flexibility, adaptability, and ability to escape local optima in search of a global solution [12].

In the context of RSA cryptanalysis, metaheuristic techniques have been applied to various aspects such as integer factorization, prime number generation, and the optimization of cryptanalytic attacks. By leveraging the strengths of these algorithms, researchers aim to improve the efficiency and effectiveness of breaking RSA encryption, potentially uncovering new vulnerabilities or optimizing existing attack methods [13]. In this paper, we use one of the most widely used optimization techniques called GAs [14].

GAs are a class of optimization algorithms inspired by the process of natural selection, as characterized by Holland [14]. GAs simulate the evolution of a population of candidate solutions through operations like selection, crossover, and mutation. These algorithms are particularly effective in solving complex optimization problems by iteratively refining solutions. Integer factorization encompasses a significant part of breaking asymmetric key ciphers like RSA [15]. We aim to explore the prime factors of large semi-prime numbers utilizing GA models. In a previous work, Mishra et al. featured a few limitations of evolutionary computation and other bio-inspired approaches for Integer factorization [15]. Notwithstanding, late effective endeavors have been made utilizing chaotic firefly algorithms, GAs, and comparable strategies [16], [17].

The authors gratefully acknowledge the financial support received from the Defence Research Development Organization (DRDO), Government of India, for this research work via Grant No. SAG/4600/TCID/CARS/C-53 as well as the support received from the Director, Scientific Analysis Group (SAG), DRDO, Delhi. The authors also acknowledge the insights provided by Prof. Nikhil R. Pal, retired Professor of Indian Statistical Institute, Kolkata during the ideation of the paper.

This paper is organized as follows: In section 2, we characterize the connected works important to figure out the paper. Thereafter, we characterize our proposed model in section 3. In section 4, we present the experimental results, in terms of success rate, number of average generations taken, running time, etc. We finally conclude the paper in section 5.

II. RELATED WORK

There are several approaches in the literature attempting to solve the Integer Factorization (IF) problem with meta-heuristic approaches. Yampolskiy et al. made one of the earliest attempts to apply GAs to the IF problem [17]. Their representation of chromosomes required the objective sets of primes, say p and q , in decimal form to deliver the number N with the end goal that $N = p \times q$. Length of the chromosome was characterized by the length of the digits in the decimal portrayal of N , with the first and second half (precisely) denoting the quantity of digits in the decimal representation of p and q , successively. The fitness function estimated the likeness of the item created by duplicating p and q from the chromosome and N utilizing parity as follows. If N was m decimal digits long, the best fitness value that could be assigned to the chromosome would be m , implying that all m digits of the product produced from the chromosome coordinated with N , and consequently the right p and q have been found. In any case, a great deal of nearby minima may likewise happen; this would occur whenever the item created by p and q from the chromosome is just a single digit off from N . The best outcome achieved by them was the factorization of a 12 digit semi-prime, requiring over 6 hrs [17].

Mishra et al. applied a multi-threaded bound-varying chaotic firefly algorithm, inspired by the behavior of fireflies, to the problem of integer factorization [18]. This algorithm employs a fitness function, with fireflies attracted to each other based on their brightness. The algorithm was tested on ten different test sets, achieving its best result with the factorization of a 14-digit semi-prime. The success rate for factoring this number exceeded 80%, depending on the number of fireflies used.

As an alternative approach, Mishra et al. used a metaheuristic algorithm based on molecules, inspired by the arrangement of a group of atoms in space where inter-atomic forces are nearly zero, to solve the IF problem [19]. This algorithm employs an energy function, a force function, and a movement function to explore potential solutions. Likewise their earlier work [18], the best result was the factorization of a 14-digit semi-prime, with a success rate of 69% to factor this number.

Recently, Rutkowski et al. applied three different approaches, two of which are GA based (with two-point crossover and tournament selection), to overcome the earlier bottlenecks. They employed mathematical knowledge concerning the distribution of primes to improve the algorithms [16]. The best one of the three GAs has a chromosome that represents m in the equation of a prime $p = 6m \pm 1$ in binary form. This approach was able to factor a number of up to 19 decimal digits. This is a significantly larger number than the largest factored by comparable methods in earlier work.

Most of the earlier approaches did not have a focus on the elitism of the GA model. In our proposed method, we have incorporated the principle of elitism and ensured that there are no duplicates in the initial population of 2000 random individuals. This enhances diversity of the population. Additionally, we made sure that the two parents selected to generate offsprings are always different, reducing redundancy—a consideration not addressed in any of the previous work.

III. PROPOSED METHOD

In the proposed model (as shown in Fig. 1), we have made modifications on one of the GA based methods suggested by Rutkowski et al. We incorporated the following important things without changing the other comparable parameters like population size and number of generations (kept as 2000 as considered in [16]).

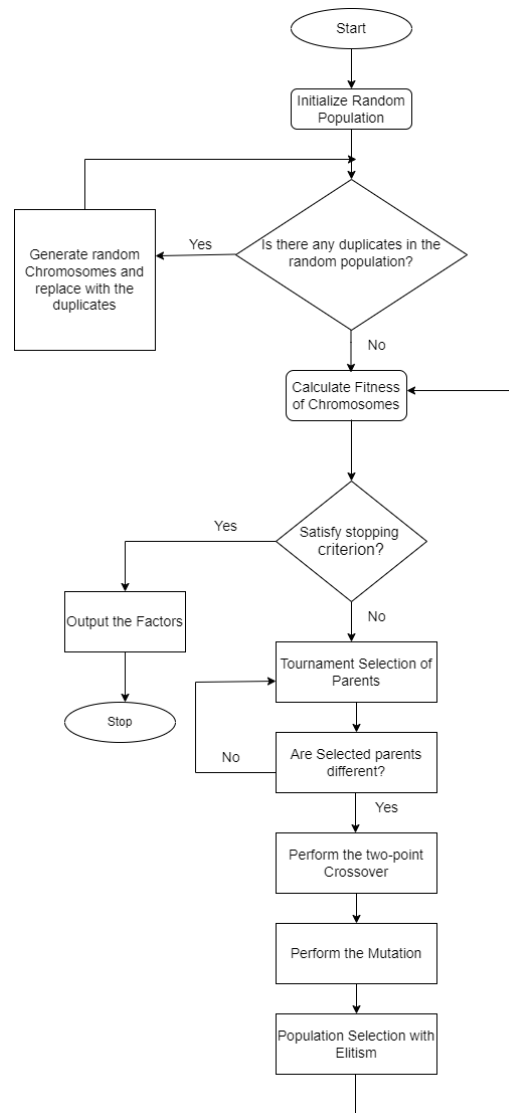


Fig. 1. Flowchart of the proposed model that incorporates elitism, ensuring there are no duplicates present in the random population size of 2000 and also making sure that the parents selected for performing the two-point crossover is always different.

- 1) **Inclusion of Elitism Principle:** Rutkowski et al. did not consider elitism in their model [16]. We have incorporated the concept of elitism by retaining elite samples between successive generations. As we have tested our algorithm with a small fraction of elitism, we hope to yield further improved results with increased elitism.
- 2) **Handling Redundancy:** To avoid redundant computations as well as to enhance diversity of the population, we have removed duplicates from the populations getting generated and also made sure that the two parents selected for generating offspring are always different.
- 3) **Diminishing the Effect of Exploitation:** We hypothesize that the Integer Factorization problem is mainly of exploration type and has hardly to do anything with the exploitation principle of evolutionary computing. Therefore, crossover operation should not have much effect on the proposed model. To verify this, we have reduced the crossover rate for improving the running time of the algorithm (refer to Fig. 1).

IV. RESULTS

We present the overall results in Tables I-III. Table I summarizes the performance of the proposed model compared to other state-of-the-art methods in terms of its ability to factorize the specified number of decimal digits. Table II summarizes the performance based on the success rate, comparing the proposed model with the state-of-the-art methods. Table III summarizes the performance based on the average number of generations required to factorize the given semi-prime number, comparing the proposed model with the state-of-the-art methods. The proposed model incorporates elitism, enabling successful factorization of numbers up to 22 decimal digits that was never achieved by any metaheuristic approach as per the literature. By eliminating redundancy, we have significantly reduced the number of generations required to reach solutions across various digit lengths in our test suite. We have also reduced the crossover rate for making the convergence faster. However, there appears to be some importance of the crossover rate for the decimal number sharing higher digits, because decreasing it to a very low value makes the model less efficient. After rigorous sensitivity analyses, we observed that our model delivers the best integer factorization results for a crossover rate of 90%, exhibiting higher success rates and fewer generations, as can be seen in Tables II and III.

The runtime for factoring a 22-digit decimal number, surpassing state-of-the-art methods, is less than an hour with 2000 generations. It is worth noting that we have also experimented with a roulette wheel selection method instead of the tournament selection method used in [16]. However, this change did not yield improved results; instead, it slowed down the algorithm's runtime.

Fig. 2 displays the variation of the best fitness scores per generation and variation of average fitness scores per generation for $N = 31,625,125,947,164,338,313$ (20 decimal digits) and $N = 3,381,755,902,745,713,031,047$ (22 decimal digits) in graphical forms. As illustrated in Figure 2, our

algorithm achieved a fitness score of "0", indicating that we found the prime factors for $N = 31,625,125,947,164,338,313$ at generation number 399 only. Similarly, Figure 3 shows that we reached a perfect fitness score of "0" for $N = 3,381,755,902,745,713,031,047$ at generation number 457.

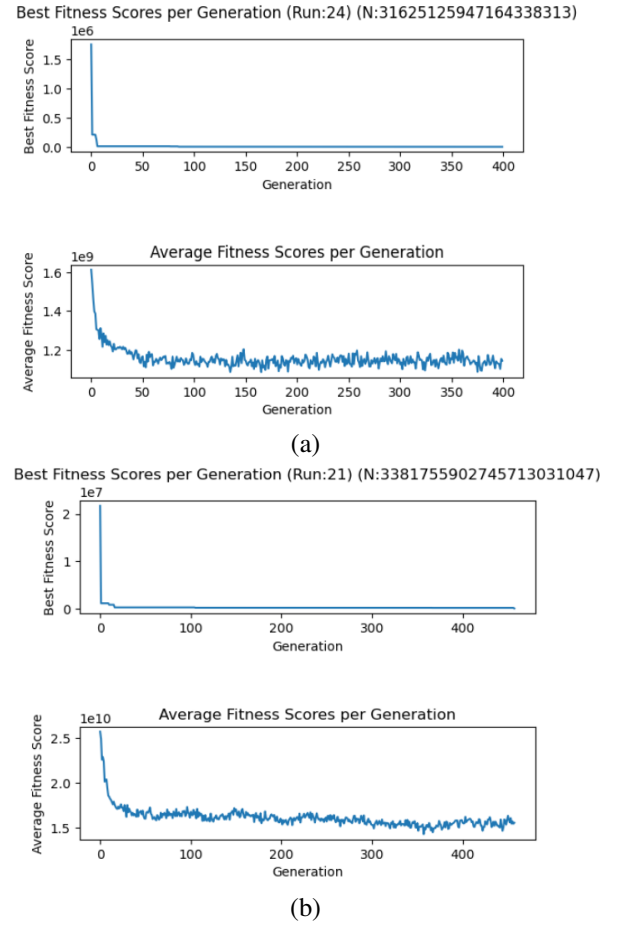


Fig. 2. (a) Average fitness score plots over successive generations for $N = 31,625,125,947,164,338,313$. (b) Average fitness score plots over successive generations for $N = 3,381,755,902,745,713,031,047$.

V. DISCUSSION

Theoretically, General Number Field Sieve (GNFS) is the fastest known approach for integer factorization [20]. However, its implementation is not scalable for larger integers. Under reasonable heuristic assumption, the asymptotic run-time complexity of GNFS is exponential. Moreover, the polynomial selection procedure, which is still a challenging thing to do, is one of the most important parts of GNFS. In addition to this, unfortunately, its complexity and overhead make it inefficient for factoring smaller integers too. So, it cannot be a general choice for factoring integers. On the other side, our proposed metaheuristic approach is based on a population based iterative calculation that incorporates independent operations under the steps fitness computation, crossover and mutation. Therefore, it can be easily parallelized on multiple clusters and scaled for larger integers. It also works well for smaller integers with a quicker convergence.

N	#Digits	Yampolskiy et al. [17]	Mishra et al. [18]	Mishra et al. [19]	Rutkowski et al. [16]	Proposed Model (Elitism + No Duplicates + 90% Crossover)
42336478013	11	Yes	Yes	Yes	Yes	Yes
272903119607	12	Yes	Yes	Yes	Yes	Yes
11683458677563	14	No	Yes	Yes	Yes	Yes
51790308404911	14	No	Yes	Yes	Yes	Yes
115137038087959	15	No	No	No	Yes	Yes
8335465900089539	16	No	No	No	Yes	Yes
10380088039872631	17	No	No	No	Yes	Yes
253422413591685001	18	No	No	No	Yes	Yes
1160633764479964633	19	No	No	No	Yes	Yes
31625125947164338313	20	No	No	No	No	Yes
454367322351811534933	21	No	No	No	No	No
3381755902745713031047	22	No	No	No	Yes	Yes
4500000514520012390279	22	No	No	No	No	No

TABLE I

SUMMARY OF PERFORMANCE (IN TERMS OF THE NUMBER OF DIGITS IN THE SEMI-PRIME FACTORIZED BY METAHEURISTICS) OF THE PROPOSED MODEL (BY INCORPORATING ELITISM, REMOVING REDUNDANCY, ADOPTING 90% CROSSOVER RATE, ALONG WITH TOURNAMENT SELECTION AND 100% MUTATION RATE AS DONE IN [16]) IN COMPARISON WITH THE STATE-OF-THE-ART METHODS. THE CASES WHERE WE ARE ABLE TO BREAK THE BOTTLENECK ARE SHOWN IN BOLD.

N	#Digits	Yampolskiy et al. [17]	Mishra et al. [18]	Mishra et al. [19]	Rutkowski et al. [16]	Proposed Model (Elitism + No Duplicates + 90% Crossover)
42336478013	11	Not Reported	Not Reported	Not Reported	30/30	30/30
272903119607	12	Not Reported	Not Reported	Not Reported	30/30	28/30
11683458677563	14	Not Reported	Not Reported	Not Reported	30/30	30/30
51790308404911	14	Not Reported	Not Reported	Not Reported	27/30	24/30
115137038087959	15	Not Reported	Not Reported	Not Reported	29/30	30/30
8335465900089539	16	Not Reported	Not Reported	Not Reported	5/30	5/30
10380088039872631	17	Not Reported	Not Reported	Not Reported	10/30	15/30
253422413591685001	18	Not Reported	Not Reported	Not Reported	2/30	6/30
1160633764479964633	19	Not Reported	Not Reported	Not Reported	1/30	1/30
31625125947164338313	20	Not Reported	Not Reported	Not Reported	-	1/30
454367322351811534933	21	Not Reported	Not Reported	Not Reported	-	-
3381755902745713031047	22	Not Reported	Not Reported	Not Reported	1/30	1/30
4500000514520012390279	22	Not Reported	Not Reported	Not Reported	-	-

TABLE II

SUMMARY OF PERFORMANCE (IN TERMS OF THE SUCCESS RATE) OF THE PROPOSED MODEL (BY INCORPORATING ELITISM, REMOVING REDUNDANCY, ADOPTING 90% CROSSOVER RATE, ALONG WITH TOURNAMENT SELECTION AND 100% MUTATION RATE AS DONE IN [16]) IN COMPARISON WITH THE STATE-OF-THE-ART METHODS. THE BEST RESULTS OVER A ROW (FOR A SPECIFIC N) ARE SHOWN IN BOLD.

N	#Digits	Yampolskiy et al. [17]	Mishra et al. [18]	Mishra et al. [19]	Rutkowski et al. [16]	Proposed Model (Elitism + No Duplicates + 90% Crossover)
42336478013	11	Not Reported	Not Reported	Not Reported	12	10
272903119607	12	Not Reported	Not Reported	Not Reported	216	557
11683458677563	14	Not Reported	Not Reported	Not Reported	415	235
51790308404911	14	Not Reported	Not Reported	Not Reported	835	686
115137038087959	15	Not Reported	Not Reported	Not Reported	548	603
8335465900089539	16	Not Reported	Not Reported	Not Reported	569	477
10380088039872631	17	Not Reported	Not Reported	Not Reported	1100	1052
253422413591685001	18	Not Reported	Not Reported	Not Reported	1440	546
1160633764479964633	19	Not Reported	Not Reported	Not Reported	1276	995
31625125947164338313	20	Not Reported	Not Reported	Not Reported	-	399
454367322351811534933	21	Not Reported	Not Reported	Not Reported	-	-
3381755902745713031047	22	Not Reported	Not Reported	Not Reported	251	457
4500000514520012390279	22	Not Reported	Not Reported	Not Reported	-	-

TABLE III

SUMMARY OF PERFORMANCE (IN TERMS OF THE NUMBER OF AVERAGE GENERATIONS TAKEN) OF THE PROPOSED MODEL (BY INCORPORATING ELITISM, REMOVING REDUNDANCY, ADOPTING 90% CROSSOVER RATE, ALONG WITH TOURNAMENT SELECTION AND 100% MUTATION RATE AS DONE IN [16]) IN COMPARISON WITH THE STATE-OF-THE-ART METHODS. THE BEST RESULTS OVER A ROW (FOR A SPECIFIC N) ARE SHOWN IN BOLD.

VI. CONCLUSION

The proposed model incorporates the combination of elitism, avoided repeated copies of initial population and parents taking part into crossover and a crossover rate of 90%. This returned us the best state-of-the-art integer factorization results, with better success rate and reduced number of generations. The running time of the said model to factor a number of 22 decimal digits (better than the state-of-the-art which was 6 hours) is less than an hour (for 2000 generations and in total of 30 complete runs). Note that the choice of roulette wheel

selection did not yield better results and in fact, it slowed down the running time of the algorithm as our fitness function is a minimization problem. In future, we plan to use recent relevant theoretical results in our model [21] and formulate the problem as a multi-objective optimization problem [22], [23].

CONFLICTS OF INTEREST/COMPETING INTERESTS

The authors declare no competing interest.

DATA AVAILABILITY

Data will be made available on reasonable request.

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